

ACCIDENTAL MICROWAVE EXPLOSION

Lessons Learned

During a routine procedure to dissolve agarose in an aqueous buffer in a microwave, the bottle of agarose exploded, blowing the microwave door off the hinges and punching a hole in the metal microwave lining. No one was injured, but the microwave was destroyed. All typical precautions were used, but additional precautions can prevent similar incidents.

What Happened?

A researcher used a 500 mL media bottle to prepare 100 mL of low-melting agarose gel with TAE buffer for electrophoresis, in which the powder was mixed and then microwaved. The cap was slightly closed, only 15-20 degrees, and the bottle was microwaved on high for two minutes using a standard domestic microwave oven. Toward the end of the 2-minute microwave heating cycle, the media bottle exploded, causing the microwave door to blow off and fall to the floor and a hole to open in the metal lining at the back of the microwave interior.

The researcher was on the other side of the lab bench, not in the direct line of the microwave, and was unharmed. Researchers in an adjacent room with the door open were closer to the microwave and were also unharmed.

Why Did This Happen?

Most likely, the agarose gel superheated in the microwave, causing a rapid expansion of gases. Either the cap was tighter than expected, or a defect in the media bottle caused a local weakness. The rapid expansion of gases overpressurized the bottle, causing it to explode with such force that the microwave was severely damaged.

What Went Right?

Several things went well, including:

1. No one was injured.
2. The researcher used typical precautions for heating agarose, including using a bottle at least 2-4 times the volume of water/gel and only slightly closing the cap.
3. The safety interlocks on the microwave worked as expected - the microwave turned off when the door came off.
4. The researcher wore appropriate personal protective equipment (PPE) for lab work, including a lab coat, safety glasses, and gloves.

How to Prevent and Respond to Accidents Like This

Heating with microwave ovens is very common in microbiology labs, and all labs need to be aware of the hazards associated with heating liquids and partially closed bottles. Common tasks that use this process include making L-agar and gel electrophoresis agar.

- Before heating in a microwave, select a bottle 3-4 times the volume of solution needed and inspect it for defects. If there are cracks, chips, or other defects, dispose of the bottle appropriately.
- Avoid heating a closed bottle, even if the lid is only slightly closed. Many applications, like electrophoresis, do not need sterile gel and, therefore, don't need to be kept closed. Heating for sterile applications like pouring plates can be done with a loose sterile cover like the bottle cap resting on top but not turned.
- Heat in short bursts, swirling the bottle intermittently (every 30 seconds) to avoid superheating and mix the solution well.
 - Stay near the microwave to observe bubbling and gauge if microwave settings need to be adjusted or the cycle is stopped early to prevent superheating and overflow.
- Add a "dummy load". Adding a second water bottle can slow heating and prevent superheating and spillage.
- Avoid the microwave - with foresight and planning, agarose bottles can be heated gently in a water bath, eliminating the risk of overheating altogether.
- Even though agarose and TAE are non-hazardous, heating them is a hazard. Wear lab attire and appropriate PPE, including a lab coat and safety glasses/goggles. Wear heat-resistant gloves when handling the heated bottle.

In case of an incident, injury, or near-miss, always report it to the PI/Lab Supervisor immediately and complete the [Incident Reporting form](#).

References

[Agarose Gel Electrophoresis for the Separation of DNA Fragments](#)

[Agarose Gel Electrophoresis](#)

[Pouring Plates from Prepared Bottled Media](#)

[Biotech Basics: Avoiding Agarose Microwave Explosions](#)